



HTTPS Configuration Guide for DAS

Welcome to the documentation for enabling HTTPS in your Data Analytics Solution (DAS).

Overview

This documentation explains how to set up HTTPS on a Data Analytics Solution (DAS) server using a PowerShell script to automate SSL certificate generation and binding. The goal is to secure HTTP communications between clients and DAS components.

Using HTTPS ensures:

- Data confidentiality and integrity
- Protection against MITM (Man-in-the-Middle) attacks
- Compliance with enterprise security standards

Why Use HTTPS?

- Encrypt communication between DAS and clients (e.g., test programs)
- Avoid security warnings from browsers and automated tools
- Prepare DAS for production environments with stricter security policies

Structure

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Prerequisites

Before executing the script, make sure the following prerequisites are fulfilled:

System Requirements

- Windows OS (Tested on Windows 10/11)
- PowerShell 5.1 or later
- .NET Framework 4.7+

Administrative Rights

- The script must be run from an elevated PowerShell session (Run as Administrator).
- Access to the **LocalMachine certificate store** is required.

Required Permissions

- Ability to execute PowerShell scripts (check with: `Get-ExecutionPolicy`)
- Ability to bind SSL certificates using `netsh`

Tips

- If running on a corporate device, Group Policies may block self-signed certificate installation. Contact IT if needed.
- Make sure the selected **port** is not already used by another application.
- Use a valid **DNS name** that matches the machine hostname or your internal network configuration.



Script Workflow

Overview

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Structure

1. Check Existing Bindings

The script first checks if the target port already has a certificate bound:

```
netsh http show sslcert
```

If the binding already exists, the script exits after displaying certificate information.

2. Create a Self-Signed Certificate

Using the `New-SelfSignedCertificate` cmdlet, a one-year self-signed certificate is generated for the specified DNS name. This is stored in the ***LocalMachine\My store***.

3. Export to Trusted Root Certification Authorities

The certificate is exported to a temporary `.cer` file and imported into the ***Trusted Root Certification Authorities*** to prevent browser warnings.

4. Retrieve the Thumbprint

The unique thumbprint of the certificate is captured to be used in the binding process.

5. Bind the Certificate to the Port

The certificate is bound to the selected port using the following command:

```
netsh http add sslcert ipport=0.0.0.0:<port> certhash=<thumbprint> appid={<guid>}
```

6. Clean Up Temporary File

The exported `.cer` file is deleted to avoid clutter.

This workflow ensures DAS can securely expose an HTTPS endpoint with minimal manual effort.

Download script

You can download the script from here [CreateAndBindCert PowerShell Script](#)



How to Use the Script

Basic Usage

To run the script, open an elevated PowerShell terminal and execute:

```
.\CreateAndBindCert.ps1 -port 443 -dnsName "localhost"
```

Parameters

Name	Type	Description
port	Integer	The TCP port number to bind the certificate.
dnsName	String	The DNS name used to create the certificate.

Example

```
.\CreateAndBindCert.ps1 -port 8443 -dnsName "das.internal.company.com"
```

This will:

- Create a self-signed certificate for das.internal.company.com
- Add it to the Trusted Root store
- Bind it to port 8443

Important Notes

- Ensure the port is not already bound to another certificate.
- If running multiple instances of DAS, use different ports or DNS names.

- Restart DAS service after the script completes for changes to take effect.



Removing an SSL Certificate Binding

If you need to unbind or clean up an existing HTTPS binding from a specific port, use the following procedure:

Command

```
netsh http delete sslcert ipport=0.0.0.0:<port>
```

Replace **<port>** with the actual port number.

Example

```
netsh http delete sslcert ipport=0.0.0.0:443
```

This command removes the SSL certificate binding from port 443.

Important

- This operation does **not** delete the certificate itself, only its binding to the port.
- If you plan to rebind a new certificate later, make sure the old one has been unbound first.
- No restart is required, but any service listening on that port over HTTPS will immediately stop working.



Security Considerations

Using HTTPS improves security, but there are still best practices to follow when working with self-signed certificates.

Use Self-Signed Certificates Only in Development

Self-signed certificates are useful for internal testing, but they are not recommended for production environments due to trust issues.

For production, use:

- Certificates signed by an internal CA
- Certificates issued by a public CA like Let's Encrypt, DigiCert, or GlobalSign

Certificate Management

- Rotate certificates before expiration (default: 1 year)
- Remove expired or unused certificates from the store
- Protect private keys with restricted access

Certificate Store Access

- Always run scripts with elevated privileges
- Ensure the certificate is stored under `LocalMachine\My`
- Add only trusted certificates to the Root CA store

Port Binding Restrictions

- Avoid binding SSL certificates to public-facing ports unless behind a firewall
- Monitor which apps are using your ports with `netstat -anob` or `Get-NetTCPConnection`

Audit & Compliance

- Keep logs of certificate creation and binding for audit purposes
- Use organization-approved tools and policies for managing certificates

These practices help ensure HTTPS connections with DAS remain secure and compliant.



Troubleshooting

Here are some common issues you may encounter when setting up HTTPS on DAS and how to fix them:

"Access Denied" when running the script

Make sure you run PowerShell as **Administrator**. Right-click on PowerShell and choose *Run as administrator*.

"Certificate not trusted" in browser

You need to import the certificate into the **Trusted Root Certification Authorities**. The script handles this automatically. If the warning persists, restart your browser or import manually.

Port already bound

Use this command to inspect current bindings:

```
netsh http show sslcert
```

Then unbind using:

```
netsh http delete sslcert ipport=0.0.0.0:<port>
```

Script doesn't create a certificate

Check if PowerShell policy blocks certificate creation. Ensure the system allows access to the certificate store and that `New-SelfSignedCertificate` is available (requires PowerShell 5.1+).

"Cannot find certificate" when binding

Make sure the thumbprint is accurate. The script uses `$cert.Thumbprint`. If you're modifying manually, ensure there are no hidden characters or typos.

DAS does not respond on HTTPS

- Ensure the DAS process is listening on the correct port.
- Verify that the firewall allows traffic on the selected port.
- Restart the DAS process after certificate binding.

When in doubt, repeat the entire process using a new DNS name and port to isolate issues.